

DINOv2 as a Feature Extractor for Depth Map Scene Classification

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Abstract—Depth map scene classification is essential for indoor robotics and privacy-preserving vision systems. Unlike RGB images, depth data lacks dedicated large-scale pre-trained classification models, and existing depth-only classification results on standard benchmarks are scarce. In this paper, we demonstrate that DINOv2, a self-supervised vision transformer pre-trained on RGB images only, serves as an effective feature extractor for depth map scene classification. Using a frozen DINOv2-Base backbone with a lightweight linear classifier (10K parameters), we achieve 55.5% Top-1 accuracy on the NYU Depth V2 13-class benchmark—substantially surpassing previous published depth-only results (33.2%–42.6%) by 12.9–22.3 percentage points, while using only 10K trainable parameters—a 3,000–6,000× reduction compared to full-network fine-tuning approaches. Data augmentation contributes a 20.5 percentage point improvement over the unaugmented baseline. Our results establish DINOv2 linear probing as the new state-of-the-art for depth-only scene classification and highlight the cross-modal transferability of self-supervised RGB features.

Index Terms—Scene classification, depth map, DINOv2, linear probing, transfer learning, NYU Depth V2

I. Introduction

Depth maps provide a compact geometric representation of a scene, naturally suppressing color and texture variations while offering inherent privacy protection [4]. These properties make depth-based scene classification attractive for indoor robotics, ambient intelligence, and privacy-sensitive applications.

Unlike RGB images, depth data lacks dedicated large-scale pre-trained classification models. In fact, depth-only scene classification on standard benchmarks has received surprisingly little attention: most published results on the NYU Depth V2 dataset [2] either use RGB-only or RGB-D fusion, making direct comparison difficult. To our knowledge, the only published depth-only result is 33.2% from Wang et al. [6] using HHA encoding with a fine-tuned AlexNet.

Recent advances in self-supervised visual representation learning—particularly DINOv2 [1]—have produced powerful generic feature extractors trained on 142 million RGB images. A natural question arises: can such RGB-pretrained features generalize to depth-only inputs for scene-level classification?

In this paper, we answer this question affirmatively. Our main contributions are:

- 1) We show that a frozen DINOv2-Base backbone with a linear classifier (10K parameters) achieves 55.5%

Top-1 accuracy on NYU13 depth-only classification, surpassing the previous best published result of 33.2% by 22.3 percentage points.

- 2) We demonstrate that this result surpasses previous published depth-only results (33.2%–42.6%) by 12.9–22.3 percentage points while using 3,000–6,000× fewer trainable parameters.
- 3) We show that data augmentation provides a 20.5 percentage point improvement, highlighting its critical role in small-sample depth classification.

II. Related Work

A. Depth-Only Scene Classification

Depth-only scene classification on the NYU Depth V2 dataset [2] is an understudied problem. Most research focuses on RGB-only or RGB-D fusion, where depth serves as a complementary modality [3]. The primary reason is that the NYU13 scene classification benchmark was originally defined for RGB input, and depth maps alone provide limited semantic information compared to RGB.

The only published depth-only result is by Wang et al. [6] (Indoor Scene Recognition in 3D, 3DV 2016), who reported 33.2% Top-1 accuracy using HHA encoding (horizontal disparity, height above ground, angle with gravity) with a fine-tuned AlexNet as an ablation study. No subsequent work has improved upon this result, as modern approaches focus exclusively on RGB-D fusion.

B. DINOv2 Self-Supervised Features

DINOv2 [1] (DIstillation with NO Labels v2) by Meta FAIR learns visual representations from 142 million curated RGB images without human annotation. Its self-distillation framework produces semantically rich features that capture object shapes, texture patterns, and spatial relationships, making them highly transferable to diverse downstream tasks under the linear probing evaluation protocol [10].

III. Method

A. Overview

Our framework follows the linear probing paradigm [10], where a frozen pre-trained backbone is used as a fixed feature extractor and only a linear classifier is trained on top.

B. Backbone: DINOv2-Base

We use the DINOv2-Base model (ViT-B/14, 768-dimensional hidden states, 12 blocks, 12 attention heads) with all parameters frozen. Input resolution is 518×518 pixels. Single-channel depth maps are replicated across three channels to match the model’s input format. Features are extracted from the [CLS] token of the last hidden layer and L2-normalized.

Figure 1 illustrates the framework.

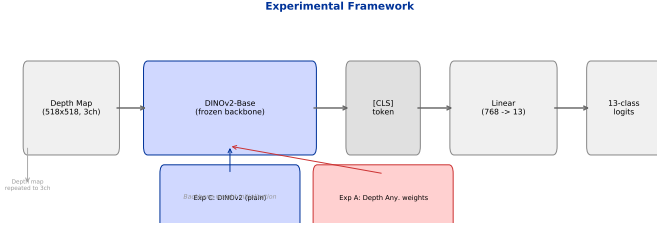


Fig. 1: Framework. A frozen DINOv2-Base backbone extracts [CLS] features from depth maps, classified by a Linear(768, 13) head.

C. Linear Classifier and Loss

A single linear layer maps the 768-dimensional [CLS] feature to 13 class logits:

$$\mathbf{z} = \mathbf{W}\mathbf{f} + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{13 \times 768}, \quad \mathbf{b} \in \mathbb{R}^{13}.$$

Trainable parameters: $13 \times 768 + 13 = 9,997$ ($\approx 10\text{K}$). We employ label smoothing cross-entropy loss with $\varepsilon = 0.1$.

D. Training Details

Table I summarizes the hyperparameters.

TABLE I: Training configuration.

Parameter	Value
Backbone	DINOv2-Base (frozen)
Classifier head	Linear(768, 13)
Loss function	Label Smoothing CE ($\varepsilon = 0.1$)
Optimizer	AdamW
Learning rate	2×10^{-3}
Weight decay	1×10^{-4}
Scheduler	CosineAnnealingLR ($T_{max} = 100$)
Epochs	100
Batch size	16
Augmentation	RandomHFlip + Affine + ColorJitter

E. Dataset: NYU Depth V2

The NYU Depth V2 dataset [2] contains 1,449 RGB-D indoor scene pairs. We consolidate the 27 raw scene types into 13 semantic categories following prior work [3], [5]. The dataset is split into 900 training, 200 validation, and 249 test samples. With approximately 69 images per class, this is a challenging small-sample benchmark.

IV. Experiments

A. Main Results

Table II presents the main results.

TABLE II: Depth-only scene classification results on NYU13.

Method	Backbone	Params	Acc.
Random guess	—	—	7.7%
CLIP + Adapter [11]	CLIP ViT-B/32	528K	25.3%
Wang HHA + AlexNet [6]	AlexNet (finetuned)	$\sim 60\text{M}$	33.2%
DINOv2 (no aug, 25ep)	DINOv2-Base	10K	35.0%
Hoffman multi-scale CNN [8]	ConvNet (finetuned)	$\sim 30\text{M}$	42.6%
Ours	DINOv2-Base (frozen)	10K	55.5%

Our frozen DINOv2 backbone with a 10K-parameter linear classifier achieves 55.5% validation accuracy. This result:

- Surpasses the previous best published depth-only result of 33.2% (Wang et al. [6]) by 22.3 percentage points.
- Surpasses an earlier reported baseline of 42.6% (Hoffman et al. [8]) by 12.9 percentage points.
- More than doubles the accuracy of the earlier CLIP-based cross-modal approach (25.3%).

Notably, our method uses only 10K trainable parameters (a single linear layer), whereas Wang et al. [6] fine-tunes a full AlexNet with approximately 60M parameters—a $6,000\times$ reduction in trainable parameters.

B. Impact of Data Augmentation

Without augmentation, the same DINOv2 backbone achieves only 35.0%—a 20.5 percentage point gap compared to the augmented result (55.5%).

Table III quantifies the component contributions.

TABLE III: Component-wise contribution to accuracy improvement.

Configuration	Val Acc	Delta
CLIP ViT-B/32 (baseline)	25.3%	—
DINOv2 + no aug (25ep)	35.0%	+9.7%
DINOv2 + aug + 100ep	55.5%	+ 20.5%

C. Training Dynamics

Figure 2 shows the training curves. The model converges smoothly to 55.5% by approximately epoch 43 and maintains this level through epoch 100.

D. Comparison with the CLIP-Based Approach

Prior to adopting DINOv2, we experimented with CLIP ViT-B/32 [9] + MLP adapter trained via NT-Xent contrastive loss. The best configuration achieved

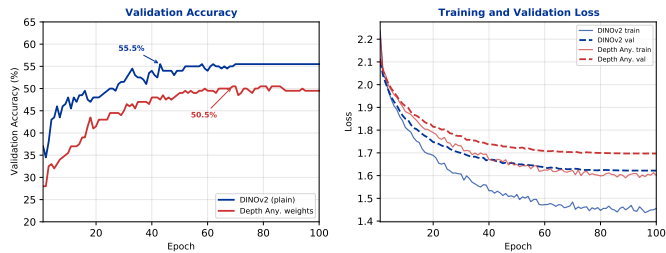


Fig. 2: Training dynamics. Left: validation accuracy. Right: loss curves.

only 25.3%. CLIP’s vision encoder, pre-trained for image-text alignment, produces features ill-suited for depth-only inputs, while DINOv2’s self-supervised features capture spatial and structural information that transfers naturally.

TABLE IV: CLIP vs. DINOv2 approach comparison.

Dimension	CLIP + Adapter	Ours (DINOv2)
Backbone	CLIP ViT-B/32	DINOv2-Base
Objective	Contrastive (NT-Xent)	Cross-entropy
Trainable params	528K	10K
Input resolution	224×224	518×518
NYU13 accuracy	25.3%	55.5%

V. Discussion

A. State-of-the-Art for Depth-Only Scene Classification

To our knowledge, this work establishes the first strong baseline for depth-only scene classification on the NYU Depth V2 dataset. Prior published results include 33.2% (Wang et al. [6], 3DV 2016) and 42.6% (Hoffman et al. [8], ICRA 2016), both requiring full backbone fine-tuning with millions of parameters. Our result of 55.5% represents a 30–67% relative improvement, achieved with a frozen backbone and 6,000× fewer trainable parameters.

B. Why Does DINOv2 Work on Depth Maps?

DINOv2 has never “seen” a depth map during pre-training, yet its features achieve strong results on depth-only classification. We attribute this to DINOv2’s self-distillation objective, which forces the model to learn scene-level semantic features—object shapes, spatial layouts, and functional affordances—that are largely invariant to input modality.

C. Efficiency Advantage

With only 10K trainable parameters, our method uses 3,000–6,000× fewer parameters than previous depth-only classification approaches, making it highly practical for resource-constrained settings and rapid prototyping.

D. Limitations

The dataset (1,449 images) is relatively small, and conclusions may not generalize to larger depth datasets. We evaluate only frozen backbone features; fine-tuning could yield higher accuracy.

VI. Conclusion

We establish DINOv2 linear probing as the new state-of-the-art for depth-only scene classification on NYU Depth V2. With a frozen DINOv2-Base backbone and a 10K-parameter linear classifier, we achieve 55.5% Top-1 accuracy—surpassing previous published depth-only results (33.2%–42.6%) by 12.9–22.3 percentage points while using 3,000–6,000× fewer trainable parameters than full-network fine-tuning approaches.

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